

Class 1.7 — Capsule Build & First Splash

Meeting date: Sunday, July 26, 2026 · 3 hours · Mr. Holcombe's house

T-MINUS STRIP

T-2 WEEKS ORDER PVC pipe (4-inch ID, 12 inches × 3-4 capsules), Schedule-40 end caps with O-ring grooves (8-12), 4-inch O-rings (Buna-N, McMaster), marine epoxy (West System 105/205 or JB Weld Marine), silicone grease.	T-1 WEEK CONFIRM lifeguard coverage IN WRITING. Confirm host pool rules sent to every parent. Verify photo releases on file for every kid expected in frame.	T-1 DAY Stage the workshop: capsule build benches (cut/sand/dry-fit station, glue station with ventilation). Pack the pool kit: capsules, towels, multimeter, watter, battery, first-aid kit, sunscreen, host thank-you note.	T-0 (MEETING DAY) Doors at 2:45 PM. Workshop half runs 3:00–4:00. Travel 4:00–4:20. Pool half 4:20–5:50. Sunscreen ALL kids before the workshop half — applied 30 min in advance for full effect.
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OWNERS AND ADVISOR DEAD-TIME ALARMS

- **Lifeguard coverage CONFIRMED IN WRITING + host pool rules to every parent + photo releases on file for every kid in frame — — · by T-1 week (HARD GATE)**
- PVC pipe (4-inch ID, 3–4 ft total) + Schedule-40 end caps (8–12) + Buna-N O-rings (4-inch, 12) — ~\$45 · by T-2 weeks
- Marine epoxy (West System 105/205 or JB Weld Marine, 8 oz) — ~\$35 · by T-2 weeks
- Silicone grease + 4-inch quartered washcloths (12) + sandpaper (220 grit) — ~\$20 · by T-1 week

STAGE IN THE ROOM

- Workshop half — capsule benches: pipe cutter, deburring sandpaper, dry-fit station, glue station (separate from cutting).
- Workshop half — ventilation: epoxy fumes need fan AND open window or door, separated from soldering bench.
- Pool kit (packed and ready by T-0): all capsules + towels + multimeter + watt meter + battery + first-aid + sunscreen + host thank-you note + camera/phone.
- Pool kit — copy of pool host's rules sheet, lifeguard contact, emergency contacts for every kid.
- Pool kit — engineering notebook for smoke test + drop log entries.

SET UP

- Workshop half: ventilation FIRST. Epoxy goes nowhere until the fan is on.
- Workshop half: gloves for the glue station — nitrile, sized small/medium/large for the kids.
- Pool half: deck setup before any kid enters water — towels stacked, kit unpacked, lifeguard introduced to the team, buddies assigned.
- Pool half: smoke-test station ON THE DECK, not in the water. Power lives on the deck.

CUE UP

- MATE "General ROV Safety" presentation (20 min) — assigned as pre-watch homework, ideally Friday or Saturday.
- Slide deck 1.7 cued on the workshop TV.
- Pool host's rules sheet printed + a copy on every parent's phone.
- Camera ready for both halves — capsule build photos AND pool-deck photos (releases verified).

STUDENTS BRING

- Swimsuit (worn under clothes for the workshop half, or in a bag).
- Towel + second towel for the bag.
- Closed-toe shoes for the workshop half; pool sandals for the deck.
- Water bottle.
- Sunscreen (apply before workshop half).
- Change of dry clothes for after.

CONFIRM · GATING

- Lifeguard coverage confirmed IN WRITING — no waiver, no maybe-the-deck-staff-counts.
- Photo releases on file for EVERY KID expected in the pool half.
- Pool host rules sent to every parent in writing.
- Buddy assignments made BEFORE any kid enters the water.
- Battery DISCONNECTED during travel and during the workshop half. Connected only on the deck under supervision.

PREP OWNERS

- Advisor — Lock lifeguard coverage in writing · by T-1 week (HARD GATE)
- Advisor — Send pool host rules to every parent · by T-1 week
- Advisor — Photo-release inventory — every kid checked · by T-1 week
- Advisor — Order PVC, caps, O-rings, marine epoxy · by T-2 weeks
- CEO (designee) — Confirm attendance + drivers for the pool · by T-1 day
- CFO (designee) — Bring engineering notebook to capture smoke test + drop log · by T-0
- Photographer — Charge camera/phone; storage capacity for 200+ photos · by T-0